THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 5: **Measurement of Length and Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A rope is 3.2 m long. After cutting off pieces of 45 cm, 55 cm and 60 cm, the remaining length is: 1

(a) 100 cm **(b)** 160 cm **(c)** 320 cm **(d)** 200 cm

2. In a pendulum experiment, 20 oscillations took 36 s. A second student gets 18 s for 10 oscillations. Both agree because: 1

(a) time period is 1.8 s in both **(b)** he is wrong
(c) 36 s vs 18 s differ **(d)** time period differs

3. A student measuring a pencil places the 2 cm mark of the scale at one end and reads 12.5 cm at the other end. Length is: 1

(a) 14.5 cm **(b)** 12.5 cm **(c)** 10.5 cm **(d)** 2.0 cm

4. A road is 2.7 km. A runner has run 1850 m. The remaining distance is: 1

(a) 1850 m **(b)** 750 m **(c)** 850 m **(d)** 650 m

5. Which of the following is true about a freely falling stone? 1

(a) it moves in a circle
(b) it is at rest
(c) it is in linear motion and its speed increases
(d) it is in periodic motion

6. A student measures a pencil three times and gets 8.3 cm, 8.2 cm and 8.4 cm. The best value to report is: 1

(a) 8.4 cm **(b)** 8.3 cm **(c)** 8.2 cm **(d)** 24.9 cm

7. A wheel of circumference 2.5 m makes 480 turns. Distance moved is: 1

(a) 1200 m **(b)** 480 m **(c)** 120 m **(d)** 2400 m

8. Which of the following shows both rotational and linear motion at the same time? 1

(a) a swing **(b)** a fan **(c)** a pendulum **(d)** a rolling ball

9. **Assertion (A):** The SI unit of length is the metre. 1
Reason (R): 1 metre is equal to 100 centimetres.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Standard units are needed to measure length. 1
Reason (R): Handspan and footstep differ from person to person.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A road runs 3.6 km. A boy walks at a step of 60 cm. How many steps does he take? 2

12. Convert 0.003 km into cm. 2

13. Why is the motion of a swing considered periodic but that of a falling leaf not? 2

14. Why can a circle's circumference not be measured with a straight ruler directly? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The circumference of a wheel is 2.2 m. How many turns will it make to cover 1.1 km? 3

16. Describe an experiment to measure the length of a curved line of 1 metre length marked on a floor by two students. 3

17. A drawing is made to a scale of 1 cm for 5 m. A building is 40 m long. What is its length in the drawing? 3

OR

A thread wound 25 times round a pencil of uniform thickness measures 8 cm in length of the coil. Find the thickness of the thread. 3

18. Explain, with two examples each, the difference between rotation and revolution. 3

OR

Why do we say that a train passenger and platform observer disagree about speed? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Sunita measured the length of her classroom using her footsteps and got 12 steps. Her friend Priya measured the same room and got 15 steps. They were confused, so their teacher in Bhopal asked them to use a measuring tape. The tape gave 9 m for both girls. 5
(i) Name the non-standard unit used by the girls.
(ii) Why did they get different values?
(iii) What is the SI unit of length?
(iv) Express 9 m in centimetres and explain why a measuring tape is better.

20. A pendulum of length 1 m makes 30 oscillations in 60 s and a pendulum of length 0.25 m makes 30 oscillations in 30 s. Calculate their time periods and compare. 5

OR

Convert and compare: in one second a snail crawls 1.2 cm and a student walks 1.2 m. Express both in cm and find how many times farther the student goes. 5

END OF QUESTION PAPER — Total: 40 Marks